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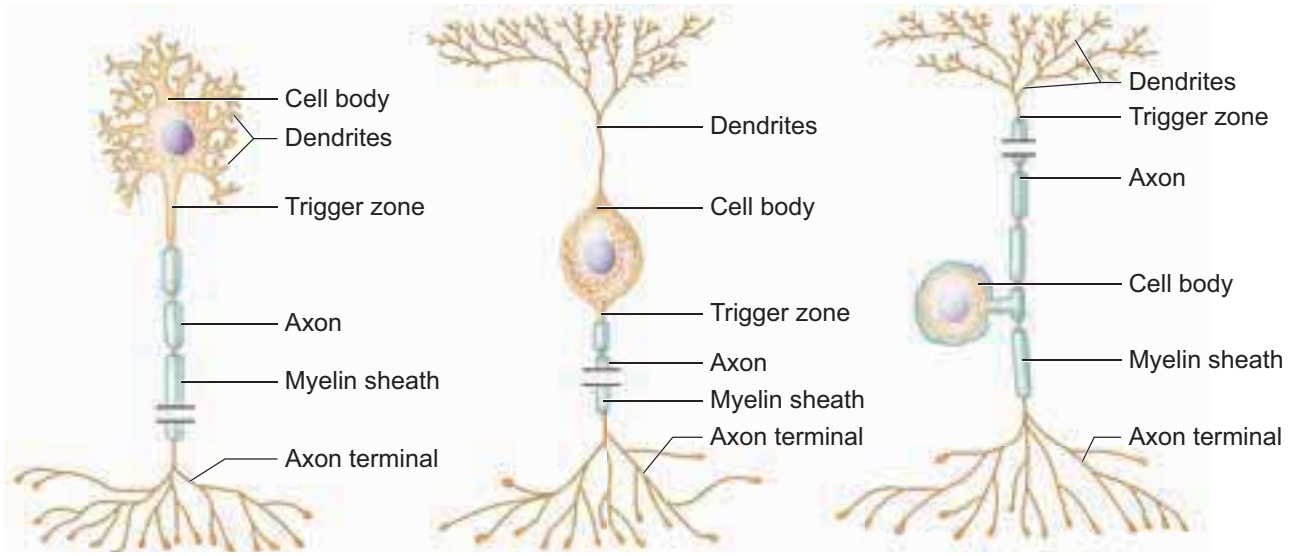


**Figure 10.2.** Types of Neurons

cells. These gaps are called **Nodes of Ranvier**. Large myelinated nerve fibres conduct impulses rapidly, whereas non-myelinated fibres conduct impulses quite slowly (Figure 10.1).

Each branch at the distal end of the axon terminates into a bulb like structure called **synaptic knob** which possesses **synaptic vesicles** filled with **neurotransmitters**. The axon transmits nerve impulses away from the cell body to an **inter neural space** or to a **neuro-muscular junction.**

The neurons are divided into three types based on number of axon and dendrites they possess (Figure.10.2).

**1. Multipolar neurons** have many processes with one axon and two or more dendrites. They are mostly interneurons.

**2. Bipolar neurons** have two processes with one axon and one dendrite. These are found in the retina of the eye, inner ear and the olfactory area of the brain.

**3. Unipolar neurons** have a single short process and one axon. Unipolar neurons are located in the ganglia of cranial and spinal nerves.

### 10.3.1 Generation and conduction of nerve impulses

This section deals with how the nerve impulses are produced and conducted in our body. Sensation felt in the sensory organs are carried by the nerve fibres in the form of electrical impulses. A nerve impulse is a series of electrical impulses, which travel along the nerve fibre. Inner to the axolemma, the cytoplasm contains the **intracellular fluid** (**ICF**) with large amounts of potassium and magnesium phosphate along with negatively charged proteins and other organic molecules.**The extra cellular fluid (ECF)** found outside the axolemma contains large amounts of sodium chloride, bicarbonates, nutrients and oxygen for the cell; and carbon dioxide and metabolic wastes released by the

**Note:** The charged particles have potential energy. The potential difference is the measure of potential energy between two points which is measured in volts or millivolts.